

CCUDA-SS: Cross-Crop Unsupervised Domain Adaptation for Semantic Segmentation

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Abstract—Accurate semantic segmentation is fundamental to precision agriculture, enabling crop coverage estimation, yield prediction, and resource optimization. However, collecting pixel-level annotations at scale is costly, motivating the use of unsupervised domain adaptation (UDA) to transfer knowledge from annotated (source) to unlabeled (target) datasets. Existing UDA methods, primarily designed for generic visual tasks, struggle with the unique domain shifts in agricultural imagery caused by variations in crop morphology, growth stage, illumination, and imaging conditions. These challenges often result in oversegmentation, under segmentation, and crop-weed misclassification. To address these issues, we propose cross-crop UDA (CCUDA), a transformer-based UDA framework tailored for agricultural domain adaptation. CCUDA leverages mid-level cues—such as row patterns and background textures—that remain consistent across crop types and environments. It employs a hybrid training strategy that combines supervised learning on the source domain with self-training on the target domain through pseudolabels. To mitigate noise in pseudolabels, we introduce masked adversarial alignment, which focusses alignment on uncertain, low-confidence regions while preserving stable representations of confident pixels. Furthermore, to resolve the feature overlap in logit space between crops and weeds due to high interclass similarity, CCUDA incorporates a logit-distance regularizer that enforces explicit separation in logit space, enhancing class discriminability. Finally, a stage-aware sampling strategy balances data exposure across varying crop growth stages, addressing dataset imbalance. CCUDA integrates seamlessly with existing UDA frameworks and, when applied to four state-of-the-art baselines across eight agricultural datasets, achieves up to +5.23% higher mean intersection over union (mIoU) and a +10.04% improvement in crop-class IoU, establishing new benchmarks for robust and generalizable agricultural segmentation.

Index Terms—Crop and weed detection, self and adversarial training, semantic segmentation, smart agriculture, unsupervised domain adaptation (UDA).

I. INTRODUCTION

NEURAL networks have achieved remarkable success across computer vision tasks, including semantic

segmentation, which is essential in precision agriculture for mapping crop coverage, detecting weeds, and monitoring growth for yield prediction and input optimization. Accurate crop coverage estimation supports decisions, such as adjusting fertilizer rates, scheduling irrigation, and predicting yield [1], [2]. However, the effectiveness of deep segmentation models depends heavily on large annotated datasets. Creating such datasets in agriculture is expensive and time-consuming because every pixel must be labeled in high-resolution images that contain dense vegetation and fine boundaries between plants and background weeds, etc. Labeling a single image can take about 2.5 h [3], making large-scale annotation infeasible [4], [5].

Even when trained on annotated data, segmentation models often generalize poorly across different fields or crop types. The visual appearance of crops varies substantially with species, growth stage, soil background, lighting, and camera setup, creating significant domain shifts between training and deployment. When applied to unseen environments, these models frequently suffer from weed–crop misclassification, where similar colour, texture, or structure causes weeds to be incorrectly identified as crops, or vice versa. This problem becomes more pronounced in ambiguous regions, such as areas where crop and weed canopies overlap or blend due to dense vegetation, shadows, or soil residue, making class boundaries visually indistinct. As a result, models often exhibit over- and under segmentation, which leads to inaccurate crop coverage estimation and consequently affects yield prediction in precision agriculture [3], [4]. Together, these challenges highlight the limitations of conventional segmentation models in addressing the complex visual ambiguity and high inter- and intraclass variability inherent to agricultural imagery. To reduce dependence on labeled data and improve transferability, unsupervised domain adaptation (UDA) adapts a model trained on a labeled source domain to an unlabeled target domain [6], [7], [8], [9]. Most UDA methods [10], [11], [12], [13] rely on self-training: the model generates pseudolabels for target images and retrain on them iteratively. While this strategy is effective in generic computer-vision tasks, it becomes unstable in agriculture, where fields vary not just visually but also biologically. When applied to crops, self-training often reinforces early mistakes—incorrect pseudolabels become “trusted” by the model in later iterations, causing predictions to drift away from the true target distribution. This gradual deterioration, termed pseudolabel drift, results in decreased model performance [14], [15], [16]. To address this limitation, adversarial alignment could be utilized; however, it results in deteriorated model performance and increased training instability [17].

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Another limitation arises from the high visual similarity between crops and weeds. Both share color, texture, and, in some cases, structure, leading their latent feature representations to overlap in the embedding space. This overlapping feature representation blurs class boundaries and increases uncertainty, causing the model to misclassify weeds as crops or vice versa. Without explicit feature regularization, self-training amplifies these overlaps, degrading adaptation quality.

A further challenge arises from the variability in crop growth stages across datasets. When the source and target domains differ significantly in their dominant phenological phases—such as early-stage crops in the source versus late-stage crops in the target—the model encounters a strong appearance mismatch. Early stages often feature small, sparse plants, while later stages exhibit dense, overlapping canopies, creating divergent visual statistics between domains. During self-training, this mismatch can bias the model toward the more frequently observed growth patterns of the source, reducing its ability to generalize across under-represented stages and leading to stage-specific performance degradation [10], [11], [12], [13]. Collectively, these challenges highlight the fundamental issues in agriculture.

Recently, UDA techniques have been extended to agriculture, particularly in precision and automated farming systems [18]. Applications include weed identification and herbicide application across fields with distinct weed species and canopy structures [19], [20], [21], crop disease detection under varying environmental and varietal conditions [22], [23], [24], and yield estimation or harvest planning across different soils, climates, and management practices [25], [26]. However, most of these methods focus primarily on architectural enhancements—such as utilizing attention mechanisms, substituting convolutional networks with transformer-based encoders, or expanding data augmentation pipelines—while overlooking the training strategies necessary to ensure stable adaptation under agriculture-specific domain shifts. As a result, these approaches often fail to resolve critical issues, such as overlapping feature representations and growth-stage variability, leading to unstable performance and limited cross-crop generalization.

To overcome these challenges, we introduce cross-crop UDA (CCUDA), a framework tailored to the agricultural domain. Built on a vision transformer (ViT) encoder-decoder backbone [13], [27], CCUDA leverages the global contextual reasoning and domain resilience of transformers to improve adaptation under strong appearance and structural variability.

CCUDA employs a self-training framework that combines supervised learning on the labeled source domain with a teacher-student scheme for generating pseudolabels on the unlabeled target domain. To tackle pseudolabel drift caused by high domain gap, CCUDA performs explicit domain alignment through adversarial learning to reduce the source-target discrepancy. However, applying adversarial alignment globally to all target pixels can introduce instability, as the discriminator may influence already well-aligned regions. To avoid this, CCUDA adopts a masked adversarial alignment mechanism that restricts adversarial supervision to low-confidence target regions, such as crop-weed boundaries, overlapping vegetation, and areas affected by illumination and viewpoint variations—where domain discrepancies are most prominent. High-confidence

regions containing stable cues, such as row structures, soil textures, and consistent background patterns, are excluded from adversarial learning to maintain reliable feature representations. This selective strategy effectively reduces noise propagation, alleviates pseudolabel drift, and stabilizes adaptation, forming the foundation of CCUDA's hybrid training (HT) strategy for robust and consistent domain generalization in agricultural imagery.

Second, to reduce logit-space ambiguity and improve class separability, CCUDA introduces logit-distance regularization (LDR). In agricultural imagery, crops and weeds often exhibit high interclass similarity and large intraclass variation, making it difficult for the model to distinguish between visually similar categories. Conventional confidence-based objectives tend to sharpen predictions without enforcing sufficient separation between class logits, leading to overlapping feature distributions and ambiguous decision boundaries. LDR addresses this by enforcing a minimum margin between the top class logits at each pixel, effectively pushing predictions away from the decision boundary. This separation encourages the model to learn fine-grained discriminative cues—such as leaf shape, edge texture, and canopy structure—thereby enhancing robustness to feature perturbations. As a result, LDR produces sharper, more stable decision boundaries and substantially reduces crop-weed confusion under domain shifts.

Third, to alleviate growth-stage imbalance, crop stage-aware sampling (SAS) is introduced, which balances exposure to different phenological phases by oversampling under-represented stages during early training. This prevents bias toward dominant growth stages and ensures stable learning across the full range of crop development.

Due to its modular design, CCUDA is fully plug-and-play: its three components, i.e., HT, LDR, and SAS, can be integrated into any UDA pipeline, making existing frameworks agriculture-aware without architectural modifications. When applied to recent UDA methods, CCUDA consistently improves performance, offering a robust and adaptable solution for agricultural domains. This compatibility is particularly valuable in precision agriculture, where most UDA advances remain focused on urban or synthetic imagery and fail to address the spatial and biological complexity of real-world crop environments. By exploiting shared structural cues and domain-consistent patterns, CCUDA bridges this gap, enabling reliable cross-crop adaptation.

Our main contributions are summarized as follows.

- 1) We identify and systematically address three major factors limiting domain adaptation in agricultural imagery—unstable pseudolabeling, feature ambiguity between crops and weeds, and bias from imbalanced crop growth stages.
- 2) We design a transformer-based UDA framework that stabilizes adaptation in uncertain regions, improves class discriminability in the logit space, and balances representation across crop growth stages. Together, these strategies establish a principled foundation for robust, annotation-efficient segmentation across diverse crops and field conditions.
- 3) We provide a comprehensive evaluation across eight agricultural datasets spanning multiple crop types and growth environments, demonstrating consistent improvements over four state-of-the-art (SOTA) UDA baselines by

up to +5.23% mean intersection over union (mIoU) and +10.04% crop-class IoU.

II. RELATED WORK

A. Semantic Segmentation

Classically, semantic segmentation models employ end-to-end CNNs based on encoder–decoder architectures, such as U-NET [28], SegNet [29], and ERFNet [30]. In these architectures, the encoder block acts as a deep semantic feature extractor, and the decoder block upsamples the features while preserving spatial information. These methods have been further improved with the concept of multiscale inputs and enhanced receptive fields at different levels of the encoder and decoder [31], [32], [33]. These architectures are widely used in agriculture for detecting and segmenting abiotic and biotic stress in crops [3], [34], [35], [36] by varying the settings of the encoder block to capture features [37]. The performance of these models has been enhanced by incorporating contextual information through multiscale attention mechanisms [38], [39]. For instance, Khan et al. [40] introduced an encoder–decoder network enhanced with channel and spatial attention units in the decoder, which improved the restoration of spatial information and enabled the localization of weeds and crops in field images. Similarly, Basalamah et al. [41] proposed an improved encoder–decoder framework that captures scene context through a global contextual module and aggregates multiscale features via a dedicated feature fusion module, enabling richer contextual understanding across varying object scales and regions of the image.

Overall, these attention and context-aggregation strategies enhance the discriminative power of CNN-based architectures in agricultural imagery. However, such models remain fundamentally locality-based and struggle to model long-range dependencies in feature maps [13], [42]. Their capacity to generalize across varying crop types, growth stages, and environmental conditions remains limited, highlighting the need for architectures that can model broader contextual relationships beyond local convolutional regions.

The evolution in attention-based modeling has, in turn, led to the introduction of ViT for semantic segmentation [42], [43]. Transformers in natural language processing tasks calculate self-attention between word tokens [44]. Similarly, ViT calculates the patch-wise attention of the image patches instead of the pixel-wise attention due to computational constraints [42]. By modeling relationships between distant patches of an image, ViTs effectively overcome the locality limitation of CNNs and capture long-range contextual dependencies within feature maps. Consequently, transformer-based architectures are now used as encoder blocks to extract features from input images [45], [46].

Recently, Cao et al. [47] implemented a pure transformer-based U-Net architecture for medical image segmentation. These pure transformer-based implementations were further enhanced by introducing control mechanisms in the attention heads [48], as multiple attention heads enable the model to capture distinct contextual relationships across spatial regions. While these advancements have gained traction in medical and

general computer vision tasks, their application in agriculture remains limited. In the context of precision agriculture, only a few studies have employed transformers for semantic segmentation tasks. Recently, SegFormer [46] and mix transformer (MiT) [49] have been used to detect anomalies in agricultural imagery [50], [51].

B. Unsupervised Domain Adaptation (UDA)

UDA methods aim to adapt a model to an unlabeled target domain. Most approaches fall into two categories: adversarial training and self-training [13]. Adversarial training aligns the source and target distributions at the input [52], feature [53], or output level. Typical strategies include style transfer with CycleGAN, pixel and feature-level alignment [53], and significance/entropy alignment [54], [55]. Although adversarial training can bridge large appearance gaps, it often suffers from training instability and mode collapse, primarily due to imbalanced discriminator–generator dynamics, high-dimensional feature spaces, and noisy or ambiguous target data that make reliable alignment difficult [13], [56].

By contrast, self-training adapts the model with high-confidence pseudolabels generated for the target domain. These pseudolabels can be obtained through confidence thresholding [57], prototype matching [58], [59], or class-balanced source–target mixing [60]. Once generated, the model is refined using these labels while applying consistency regularization across data augmentations, image crops, or ensemble predictions to maintain stable outputs under perturbations [61], [62], [63]. Recent work further boosts robustness by coupling strong augmentations with multiview or multiteacher ensembles [64], [65], [66]. Building on this direction, Asad et al. [64] proposed an ensemble-based UDA framework for agricultural imagery. They combined several crop and weed segmentation models trained on different datasets within a teacher–student paradigm, where the ensemble of base models acted as *teachers* to generate pseudolabels. The final predictions being refined using a trainable meta-network to fuse outputs from the teachers, enabling the model to integrate complementary information from multiple experts and adapt effectively to heterogeneous crop and field conditions.

While generally more stable than pure adversarial schemes [11], self-training-based architectures are prone to pseudolabel drift and confirmation bias [10], [12], [13]. These issues intensify under strong agricultural domain shifts, as agricultural imagery exhibits extreme variability driven by biological and environmental factors—specifically, high intraclass variation and high interclass similarity between crops and weeds. Both share similar colour, texture, and structure; their appearance changes dynamically across growth stages; and imaging conditions vary with soil background, viewpoint, and illumination. These issues cause early pseudolabels to be noisy and self-reinforce errors, leading to unstable adaptation. While adversarial alignment could alleviate this issue, existing works [10], [13], [17] avoid it due to instability in high-resolution settings. We argue that the instability of adversarial alignment in high-resolution settings arises from

applying adversarial objectives indiscriminately to all pixels, regardless of model confidence. This global alignment disrupts already well-aligned, high-confidence regions—i.e., stable features—and introduces unnecessary noise. In contrast, the true domain gap lies in low-confidence pixels, such as crop–weed boundaries and overlapping vegetation, where ambiguity and biological variability are greatest.

To address this, CCUDA introduces masked adversarial alignment, which selectively aligns uncertain regions while preserving domain-invariant cues. High-confidence pixels, typically representing consistent areas, such as soil, residue, and well-learned crop zones, are excluded from adversarial supervision to prevent the corruption of reliable features. By utilizing HT with masked adversarial learning focussing only on ambiguous regions and leaving confident areas intact, CCUDA stabilizes self-training, suppresses pseudolabel drift, and achieves robust domain alignment without compromising the integrity of stable features.

Moreover, most existing UDA frameworks complement self-training with entropy minimization [55], [67], [68] to encourage confident predictions in the target domain. While effective in domains with well-separated classes (e.g., urban scenes), this strategy performs poorly for agricultural imagery, where crops and weeds exhibit high interclass similarity and large intraclass variation. By sharpening prediction probabilities, entropy minimization reduces uncertainty but fails to constrain the relative geometry of decision boundaries in feature space. Consequently, when feature representations of visually similar classes overlap in logit space—such as crops and weeds—the model can produce high-confidence yet semantically ambiguous predictions, thereby weakening its discriminative capacity. This issue is aggravated when weeds are annotated as background [64], since the network’s attention spreads over soil, residue, and shadows, further blurring class separation and amplifying pseudolabel drift [12], [13], [69].

To overcome this limitation, CCUDA introduces LDR, which enforces a minimum margin between class logits at each pixel. By maximizing this margin, predictions are pushed away from decision boundaries, enhancing robustness to feature perturbations and promoting stable, discriminative separation in the logit space. Unlike entropy minimization, which encourages uniform confidence, the logit-distance constraint induces selective confidence by explicitly forcing one class to dominate. This yields sharper, more reliable decision boundaries and substantially reduces crop–weed confusion under domain shifts.

Long-tailed label distributions pose another persistent challenge in UDA, as models tend to bias toward frequently occurring classes, leading to the under segmentation of rare ones. Existing methods address this through rare class sampling (RCS) [17], [70], which rebalances semantic classes by oversampling under represented categories. However, this mechanism is ill-suited for agricultural imagery, where imbalance arises within a single class across phenological stages, rather than between distinct classes. As the same crop appears vastly different across growth stages, it causes stage-specific under representation that RCS fails to correct.

CCUDA addresses this fundamental limitation by introducing SAS, a stage-sensitive sampling strategy that balances the representation of crops across their lifecycle. SAS clusters images based on crop-coverage ratios and selectively oversamples under represented growth stages, thereby reducing confirmation bias and enhancing generalization across phenological variations. Unlike heuristic resampling or class weighting used in plant phenotyping [71], SAS operates adaptively without additional hyperparameters, ensuring consistent segmentation performance throughout different growth stages.

III. METHODOLOGY

We propose domain-robust training strategies to mitigate the difficulties that UDA methods face in precision agriculture scenarios. We observe that different fields share visually consistent background cues, including inter-row soil, crop residue, and common weeds, which change little between locations and are abundant in both the source and target data. Using this insight, we define a baseline HT approach that addresses the challenges of domain shift and changes in crop shapes. We then apply LDR to HT; By enlarging the margin between the two highest logits at each pixel, it sharpens the crop–background decision boundary and thereby reduces weed-as-crop misclassification when weeds visually blend with or occlude crop plants. The overall loss function for training consists of self-training loss, *masked adversarial* loss, and LDR loss, given by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{self}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{ldr}}. \quad (1)$$

Finally, we introduce a novel SAS algorithm to combat crop-stage bias. It emphasizes the representational balance of the crop stages, ensuring equitable learning and performance across all crop stages. The flow diagram of the proposed framework is shown in Fig. 1.

A. HT for UDA

A neural network denoted as g_θ is initially trained on the source domain crop using a supervised loss function. To address the issue of UDA, where we only have access to source domain labels, we adopt a teacher–student-based architecture using a teacher network h_ϕ to generate pseudolabels for target crop (target domain) images, which are then used as ground truth to train the student network g_θ . However, if the teacher assigns incorrect labels, those errors are feedback to the student; the updated student is then averaged into the teacher, which in turn generates the next round of pseudolabels. This feedback loop can amplify early mistakes, leading to pseudolabel drift and cumulative error propagation [10]. To tackle these issues and improve domain invariance within the primary network (g_θ), adversarial training [55] is considered. However, plain adversarial training is notoriously unstable when source and target crops differ markedly in morphology and growth stage [64], [72]. Therefore, we reinforce the masked adversarial training, applying the adversarial objective *only* to low-confidence (uncertain) pixels. By allowing the discriminator to inspect only ambiguous regions—those most affected by domain shift, such as disparate crop morphologies, varying illumination, and oblique imaging

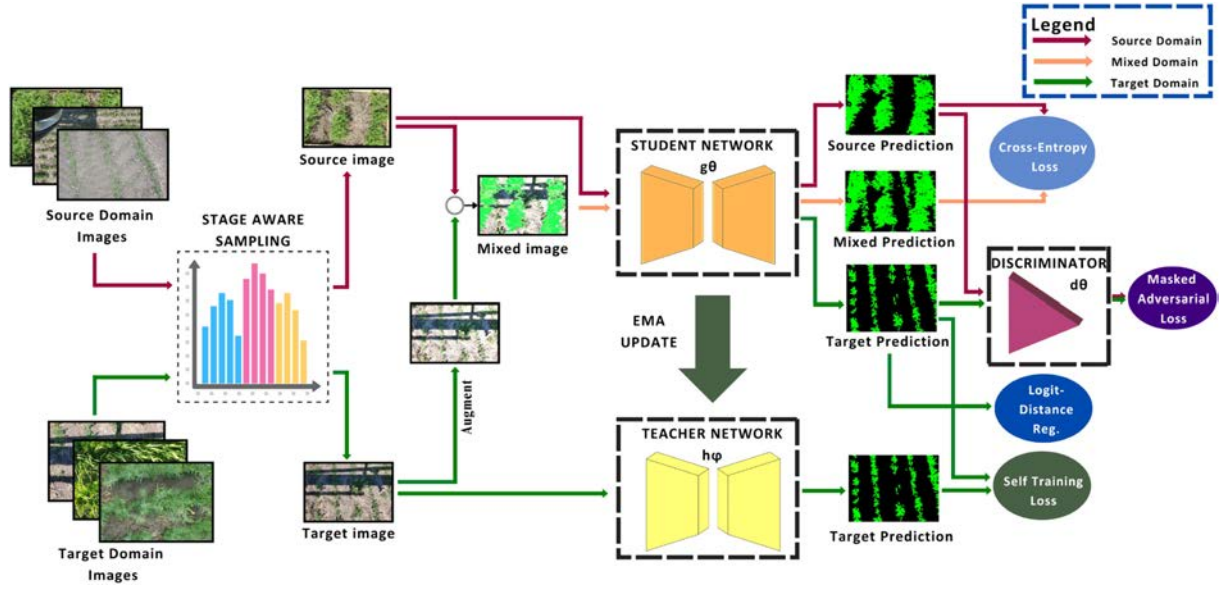


Fig. 1. Proposed CCUDA framework begins with the SAS block, which selects and augments input images before feeding them into the student network (g_θ). The student's predictions on the target domain are refined by a teacher network (h_ϕ), updated via EMA to provide stable supervision. A discriminator (d_θ) applies a masked adversarial loss to align source and target distributions, focusing specifically on low-confidence regions. In addition, LDR is applied to the target domain's prediction from the student's output, encouraging larger logit margins between crop and background classes, thereby reducing crop–weed confusion. A self-training loss on high-confidence pseudolabels further enhances target domain adaptation.

angles—we avoid the noisy gradients and mode collapse that can occur in the case of full-map adversarial learning [10]. Thus, it achieves greater training stability and stronger cross-crop adaptation, as evidenced in Table IV.

1) *Self-Training-Based Pseudolabel Generation*: The student model, g_θ , is provided with source crop images $X_S = \{x_S^{(i)}\}_{i=1}^{N_S}$ and their labels $Y_S = \{y_S^{(i)}\}_{i=1}^{N_S}$, along with mixed crop/domain images (source and target images fused) $X_M = \{x_M^{(i)}\}_{i=1}^{N_M}$ and their labels $Y_M = \{y_M^{(i)}\}_{i=1}^{N_M}$. Each image $x \in \mathbb{R}^{H \times W \times 3}$ has a height H and a width W , with each dataset containing C semantic classes. The target crop pseudolabels from h_ϕ fused with the source labels are used as mixed-crop image labels. Thus, the self-training loss ($\mathcal{L}_{\text{self}}$) is given by

$$\mathcal{L}_{\text{self}} = \mathcal{L}_{S+M} + \lambda_T \mathcal{L}_T. \quad (2)$$

Here, $\mathcal{L}_{S+M}$ refers to the categorical cross-entropy loss calculated for direct segmentation using both y_S and y_M , with λ_M being the weighing factor for the mixed crop segmentation loss

$$\begin{aligned} \mathcal{L}_{S+M} = & - \sum_i \left(\sum_{j=1}^{H \times W} \sum_{c=1}^C y_{S(j,c)}^{(i)} \log g_\theta \left(x_{S(j,c)}^{(i)} \right) \right. \\ & \left. + \lambda_M \sum_{j=1}^{H \times W} \sum_{c=1}^C y_{M(j,c)}^{(i)} \log g_\theta \left(x_{M(j,c)}^{(i)} \right) \right). \end{aligned} \quad (3)$$

The term $\mathcal{L}_T$ further trains g_θ using target crop images $X_T = \{x_T^{(i)}\}_{i=1}^{N_T}$

$$\begin{aligned} \mathcal{L}_T = & - \sum_i \sum_{j=1}^{H \times W} \sum_{c=1}^C q_T^{(i)} \mathbb{1} \left[c = \arg \max_{c'} h_\phi(x_T^{(i)})_{(j,c')} \right] \\ & \cdot \log g_\theta(x_T^{(i)})_{(j,c)}. \end{aligned} \quad (4)$$

Since pseudolabels in the target domain may be noisy or incorrect, particularly under severe domain shifts, we adopt a pseudolabel selection mechanism to mitigate error propagation. Specifically, we use a confidence estimate $q_T^{(i)}$, following [10], [12], to weight the contribution of each image based on the reliability of its pixel-level pseudolabels. For semantic segmentation, the quality estimate is computed as the ratio of pixels exceeding a confidence threshold τ of the maximum softmax probability

$$q_T^{(i)} = \frac{\sum_{j=1}^{H \times W} \left[\max_{c'} h_\phi(x_T^{(i)})_{j,c'} > \tau \right]}{H \cdot W}. \quad (5)$$

Here, $h_\phi(x_T^{(i)})$ denotes the softmax-normalized teacher predictions for the target image $x_T^{(i)}$. The confidence map $q_T^{(i)} \in [0, 1]^{H \times W}$ scales the target loss ($\mathcal{L}_T$), giving more weight to reliable predictions while down-weighting uncertain ones. The threshold τ defines the confidence level above which pixels are treated as valid pseudolabels. A higher τ improves label precision but reduces spatial coverage, whereas a lower τ increases coverage at the risk of including incorrect labels. This adaptive weighting mitigates noisy gradients, limits pseudolabel drift, and stabilizes training under strong domain shifts.

Subsequently, at each iteration t , the teacher weights (ϕ_t) are updated via the exponential moving average (EMA) of the student weights (θ_t) with a decay factor α

$$\phi_{t+1} \leftarrow \alpha \phi_t + (1 - \alpha) \theta_t. \quad (6)$$

2) *Masked Adversarial Training for Uncertain Regions*: Only self-training deployment suffers from two persistent shortcomings: (i) low-confidence pixels pass unreliable pseudolabels forward, and (ii) the domain gap is concentrated in those same ambiguous pixels [64]. We remedy both issues by exposing only low-confidence pixels to a discriminator. Pixels already

predicted with high confidence, mainly row-aligned soil, crop residue, vegetation, and background weeds whose visual appearance changes little across fields, are ignored. Therefore, the discriminator focusses on pixels that are hard to classify, where morphology, growth stage, imaging angle, and illumination differ the most between domains.

Given a prediction map $g_\theta(x) \in \mathbb{R}^{H \times W \times C}$, per-pixel confidence is defined as

$$\pi_{(j)} = \max_c \text{softmax}(g_\theta(x))_{(j,c)}, j \in 1, \dots, H \cdot W \quad (7)$$

and a binary mask

$$\text{mask}(g_\theta(x)) = \{j \mid \pi_{(j)} < \beta\}, \quad \beta \in (0, 1). \quad (8)$$

It selects pixels whose highest softmax score falls below a threshold β . The discriminator d_θ receives only these masked regions. Its loss is given as

$$\begin{aligned} \mathcal{L}_D = \min_{\theta} & \left(\frac{1}{N_S} \sum_i \mathcal{L}_{\text{CE}} \left(d_\theta \left(\text{mask} \left(g_\theta \left(x_S^{(i)} \right) \right) \right), 1 \right) \right. \\ & \left. + \frac{1}{N_T} \sum_i \mathcal{L}_{\text{CE}} \left(d_\theta \left(\text{mask} \left(g_\theta \left(x_T^{(i)} \right) \right) \right), 0 \right) \right). \quad (9) \end{aligned}$$

Conversely, the generator (student) minimizes

$$\mathcal{L}_{\text{adv}} = \min_{\theta} \lambda_{\text{adv}} \frac{1}{N_T} \sum_i \mathcal{L}_{\text{CE}} \left(d_\theta \left(\text{mask} \left(g_\theta \left(x_T^{(i)} \right) \right) \right), 1 \right) \quad (10)$$

encouraging low-confidence target pixels to appear indistinguishable from source ones. Pixels already predicted with high confidence ($\pi_{(j)} \geq \beta$) are excluded, allowing the discriminator to specialize in hard examples while the student receives focused gradients that precisely refine those regions. Weighted by λ_{adv} , the masked adversarial terms are integrated with the supervised, self-training, and LDR components, resulting in a stable hybrid objective.

B. Enhanced HT With LDR

Segmentation on unlabeled target images frequently plateaus on pixels whose class scores are narrowly separated, leading to ambiguous pseudolabels and slower domain alignment. To address this ambiguity, we introduce LDR, which explicitly enforces a minimum margin between the two highest class scores at every pixel. Applied to the student network's predictions on target images, this term produces sharper decisions, provides higher quality pseudolabels, and propagates improved confidence to the teacher through the EMA update.

For a target image $x_T^{(i)}$ and a pixel location j , let the raw class scores (logits) be $z_{(i,j)} \in \mathbb{R}^C$. The top and second top logit are given by

$$z_{(i,j)}^{(1)} = \max_c z_{(i,j,c)}, \quad z_{(i,j)}^{(2)} = \max_{c \neq \arg \max_{c'} z_{(i,j,c')}} z_{(i,j,c')}.$$

The LDR loss is given as

$$\mathcal{L}_{\text{ld}} = \lambda_{\text{ld}} \frac{1}{N_T H W} \sum_{i=1}^{N_T} \sum_{j=1}^{H \times W} \max \left(0, m - \left(z_{(i,j)}^{(1)} - z_{(i,j)}^{(2)} \right) \right) \quad (11)$$

where m is a margin hyperparameter. Pixels with a score gap below m incur a hinge penalty; confident pixels (gap $\geq m$) contribute zero, allowing the network to focus on the ambiguous subset. The term $\mathcal{L}_{\text{ld}}$, weighted by λ_{ld} , is incorporated into the overall objective, thus tightening the decision margins.

C. Stage Aware Sampling

Model bias toward dominant classes is a persistent problem in most UDA methods. As the model processes images, it inherently learns features of the dominant class, leading to the misclassification of less-represented classes. This issue is compounded by self-training, as the teacher model inadvertently reinforces the bias [10]. In the agricultural domain, where typically only two classes exist, i.e., crop and background, with several growth stages of crops, i.e., early, mid, and late, addressing this bias within a single class, i.e., crop, is crucial. These developmental stages of crops play a critical role in the agricultural domain. When late-stage crop images are predominantly passed through the model, it becomes biased toward later stages, diminishing performance on early-stage images and vice versa. This issue is exacerbated when the source domain dataset contains an overwhelming number of one-stage compared to other crop stages. To address this, we propose a novel SAS algorithm that preferentially selects the least and moderately represented crop stages in the dataset more frequently than the dominant stage. The SAS algorithm processes images to analyse crop percentages and clusters them into different growth stages based on these percentages. Initially, it calculates the crop percentage $P_{\text{crop}}(i)$ for each image i as given in the following equation:

$$P_{\text{crop}}(i) = \left(\frac{\text{crop pixels in } i}{\text{total pixels in } i} \right) \times 100. \quad (12)$$

These percentages are scaled using $P_{\text{crop scaled}}(i) = \frac{P_{\text{crop}}(i) - \mu}{\sigma}$ where μ and σ are the mean and standard deviation of P_{crop} . Then, K -means clustering is applied to $P_{\text{crop scaled}}$ to classify each image into one of a total of three ("early," "mid," and "late") growth stages.

After clustering, each image is assigned a uniform initial probability $P_{\text{init}}(i) = \frac{1}{N}$, where N is the total number of images present in the dataset. To ensure equity in sampling, an equity factor E_s is computed for each stage s , as given in the following equation:

$$E_s = \frac{P'(s)}{P_{\text{obs}}(s)} \quad (13)$$

where P' is the ideal distribution representing the uniform distribution of images across the three stages ("early," "mid," "late"). Each stage should ideally have an equal proportion of images, making P' a vector of three elements, each with a value of $\frac{1}{3}$. Whereas P_{obs} is the observed distribution, representing the observed distribution of images in each stage after clustering. It is also a vector, where each element corresponds to the proportion of images assigned to one of the stages. After calculating the equity factor E_s for each stage s , we multiply it with the initial probabilities $P_{\text{init}}(j)$ of all images j in stage s , giving

each image in that stage the same probability

$$P_{\text{stage}}(s) = \sum_{j \in s} P_{\text{init}}(j) \times E_s.$$

Then, we reassign this total probability of the stage to each image i within the stage s based on the crop percentage $P_{\text{crop}}(i)$ of each image. The final probability $P_{\text{final}}(i)$ for each image belonging to a specific stage s is given by

$$P_{\text{final}}(i) = P_{\text{stage}}(s) \times \frac{P_{\text{crop}}(i)}{\sum_{k \in s} P_{\text{crop}}(k)}.$$

This ensures that the final probabilities are normalized so that their sum across all images equals one. The pseudocode for SAS is given in Algorithm 1. SAS provides a fair representation of each growth stage in the analysis, helping to balance out any inherent sampling biases. The stages are determined by the outputs from the K -means clustering rather than hard thresholding, allowing the algorithm to classify the images objectively based on the data. This approach enables the algorithm to be applied to other datasets with many classes, addressing the issue of lower accuracy on rare classes.

IV. RESULTS

A. Implementation Details

Datasets: This study utilizes eight datasets from four different crops: maize, lentils, flax, and narrow leaf. We utilize public datasets from the ROSE challenge [73], which consist of maize and bean datasets from four agriculture robot-based teams, namely BIPBIP, PEAD, ROSEAU, and WeedElec. Each dataset was collected using a different robot and camera system, leading to significant domain gaps in the images due to variations in setup and agricultural field conditions. The BIPBIP team collected datasets in both 2019 (early crop growth stage) and 2021 (late crop growth stage), highlighting significant differences in crop growth stages. The rose challenge datasets comprise a total of 1000 labeled RGB images, with 625 images belonging to the maize crop. The image resolutions for the datasets collected by BIPBIP and WeedElec remain at 2048×1536 and 5184×3456 , respectively.

In addition, we introduce four new datasets: the lentil crop dataset (LCD), the flax crop dataset (FCD), the narrow-leaf crop dataset (NCD), and the maize crop dataset (MCD). These datasets were collected in 2022 across fields in the Canadian Prairies by Croptimistic Technology Inc. [74], using high-resolution RGB imagery captured with Samsung S10 [75] smartphone cameras mounted on ground-based agricultural machinery, such as sprayers and seeders. The images, originally intended for crop establishment and weed monitoring, were acquired in a grid pattern (70×80 feet) without overlap, resulting in a total of 1287 RGB images, each with a resolution of 1440×1080 pixels. Each image typically captures 3–5 crop rows in uncontrolled natural field environments.

The LCD dataset comprises 392 images, all of which were taken under clear, sunny weather conditions. The FCD dataset includes 234 images that contain shadows of equipment. The NCD dataset consists of 236 images, primarily captured at night. The MCD dataset contains 425 images that feature blurred

Algorithm 1: Stage Aware Sampling.

Require: Set of images D

Ensure: Final probabilities $P_{\text{final}}(i)$ for each image i

- 1: Initialise $N \leftarrow$ number of images, $S \leftarrow 3$ // Number of growth stages ("early", "mid", "late")
- 2: **for** each image $i \in D$ **do**
- 3: Compute $P_{\text{crop}}(i) = (\frac{\text{crop pixels in } i}{\text{total pixels in } i}) \times 100$
- 4: **end for**
- 5: Compute the mean μ and standard deviation σ of P_{crop} :

$$\mu \leftarrow \frac{1}{N} \sum_{i=1}^N P_{\text{crop}}(i)$$

$$\sigma \leftarrow \sqrt{\frac{1}{N} \sum_{i=1}^N (P_{\text{crop}}(i) - \mu)^2}$$

- 6: **for** each image $i \in D$ **do**
- 7: Scale the crop percentages $P_{\text{crop scaled}}(i) = \frac{P_{\text{crop}}(i) - \mu}{\sigma}$
- 8: **end for**
- 9: Initialize $k = 3$ centroids μ_1, μ_2, μ_3 randomly
- 10: **repeat**
- 11: **for** each image $i \in D$ **do**
- 12: Assign image to nearest centroid:

$$s(i) = \arg \min_{j \in \{1,2,3\}} |P_{\text{crop scaled}}(i) - \mu_j|$$

- 13: **end for**
- 14: **for** each centroid μ_j **do**
- 15: Update centroid:

$$\mu_j = \frac{\sum_{i:s(i)=j} P_{\text{crop scaled}}(i)}{\sum_{i:s(i)=j} 1}$$

- 16: **end for**
- 17: **until** convergence
- 18: **for** each image $i \in D$ **do**
- 19: Assign initial probability:

$$P_{\text{init}}(i) = \frac{1}{N}$$

- 20: **end for**
- 21: **for** each stage $s \in \{\text{early, mid, late}\}$ **do**
- 22: Compute ideal distribution $P'(s) \leftarrow \frac{1}{3}$
- 23: Compute observed distribution
- 24: $P_{\text{obs}}(s) \leftarrow \frac{\text{number of images in stage } s}{N}$
- 25: Compute equity factor $E_s \leftarrow \frac{P'(s)}{P_{\text{obs}}(s)}$
- 26: **end for**
- 27: **for** each stage s **do**
- 28: Compute stage probability:

$$P_{\text{stage}}(s) = \sum_{j \in s} P_{\text{init}}(j) \times E_s$$

- 29: **end for**
- 30: **for** each stage s **do**
- 31: Reassign probabilities:

$$P_{\text{final}}(i) = P_{\text{stage}}(s) \times \frac{P_{\text{crop}}(i)}{\sum_{k \in s} P_{\text{crop}}(k)}$$

- 32: **end for**
- 33: **end for**
- 34: Normalise P_{final} so that:

$$\sum_{i=1}^N P_{\text{final}}(i) = 1$$

- 35: **return** $P_{\text{final}}(i)$
-

visuals and equipment shadows. These conditions collectively represent various challenging domains commonly encountered in agricultural fields. For annotations, all images in the new datasets are labeled using segments.ai [76]. Crops are annotated as the crop class, whereas pixels containing crop residue, weeds, and soil are labeled as background.

Following the protocol in [64], each target domain dataset is split into 70% training and 30% validation. All results are reported on the validation set.

Network architecture: Our network is based on the MM-Segmentation framework [78]. DAFormer [13] serves as the base architecture, employing MiT-B5 [46] as the encoder block initialized with *ImageNet-pretrained weights* for robust visual representation learning and an atrous spatial pyramid pooling decode head for context-aware feature fusion. CCUDA is designed as a plug-and-play framework and is integrated on top of MIC [12] to ensure direct and fair comparison with existing UDA methods. The masked adversarial alignment operates on output logits from the decoder, while LDR is applied in the logit space to enhance class separability. The discriminator follows the ADVENT design [55], consisting of convolutional layers with leaky rectified linear unit (ReLU) and instance normalization to distinguish source and target predictions.

Training: In the default UDA setting, we follow the multiresolution self-training strategy of high-resolution domain adaptation (HRDA) [10] and MIC [12], along with their training parameters. Both the encoder and decoder of the model (g_θ) are optimized using AdamW [79] with learning rates of 6×10^{-5} and 6×10^{-4} , respectively, and a linear warm-up for the first 1.5k iterations followed by polynomial decay. The model is trained for 7k iterations with a batch size of 2 (one source and one target image per iteration). Following the multiresolution protocol of [10], [12], full-resolution images are used for training. Data augmentation follows domain adaptation via cross-domain mixed sampling (DACS) [60], including random cropping, horizontal flipping, Gaussian blur, and colour jitter. The mixed domain images are formed using [10]. The teacher network (h_ϕ) shares the same architecture as g_θ and is updated by an EMA with a decay factor $\alpha = 0.999$, the pseudolabel confidence threshold $\tau = 0.9$, and weighting factors $\lambda_M, \lambda_T = 0.01$.

For masked adversarial training, the discriminator is optimized using AdamW [79] with a learning rate 1×10^{-4} and masking threshold of $\beta = 0.9$. The $\mathcal{L}_{\text{adv}}$ applied to the segmentation model is weighted by $\lambda_{\text{adv}} = 0.001$, encouraging stable domain alignment. Only low-confidence target pixels (confidence $\pi < \beta$) are selected for adversarial updates to focus adaptation on uncertain regions. LDR is applied to target-domain predictions with $\lambda_{\text{ld}} = 0.01$ and a margin $m = 1.5$ (see Table IX). For SAS, under represented crop stages are oversampled early in training to maintain balanced exposure across phenological phases.

Since CCUDA is a plug-and-play framework, it inherits other loss components from MIC [12]. All experiments are conducted on a NVIDIA GeForce RTX 3090 GPU (24 GB) for 7 k iterations. Evaluation is performed on full-resolution target-domain images using class-wise IoU, mIoU, accuracy (%), precision

(%), recall (%), and average precision (AP), with the oracle (see Table I) representing the upper bound obtained using supervised learning on the target with its labels. All reported results are averaged over three data sampling seeds.

B. Comparison With the SOTA UDA Methods

Table I provides a comprehensive comparison of our proposed CCUDA method against existing SOTA UDA techniques across six challenging agriculture-specific source-to-target adaptations. CCUDA consistently outperforms previous UDA methods by substantial margins, achieving improvements of approximately 4.49% to 10.9% in IoU-Crop and 1.29% to 5.23% in mIoU. These significant improvements confirm the efficacy of CCUDA in addressing domain-specific shifts in agriculture.

Each evaluated source-to-target scenario in Table I captures critical shifts in the agricultural domain previously identified. In particular, CCUDA effectively handles variations in crop growth stages, especially in scenarios like LCD $\rightarrow$ FCD, where adaptation is required from predominantly late-stage lentil images (source) to early-stage flax images (target). Furthermore, CCUDA robustly mitigates extreme imaging conditions' shifts, such as night-to-daytime transitions (NCD $\rightarrow$ WeedElec Maize), normal-to-blurred visuals (NCD $\rightarrow$ MCD), and significant morphological differences between narrow-leaf and MCD pairs, as in (BIPBIP-2021-Maize $\rightarrow$ NCD and NCD $\rightarrow$ MCD).

These performance gains are further reinforced by the qualitative comparisons shown in Fig. 2. Existing UDA methods often encounter issues, such as pseudolabel drift, resulting in confusion between weeds and crops and oversegmentation in complex agricultural scenes. CCUDA notably reduces these misclassifications by learning sharper and more discriminative decision boundaries, which is clearly evident in PEAD Maize $\rightarrow$ NCD and NCD $\rightarrow$ WeedElec Maize adaptations, where it eliminates weed-as-crop misclassification despite the severe domain gap due to imaging setup (camera angle and lighting) and crop morphology. In addition, CCUDA efficiently addresses over segmentation issues, especially in blurred and ambiguous regions, such as those encountered in the NCD $\rightarrow$ BIPBIP-2019-Maize scenario, where MIC and HRDA capture a robot stand as a crop. Furthermore, despite challenging night-to-day blurred scenarios, such as NCD $\rightarrow$ MCD, our method avoids making noisy predictions. Morphological variability caused by growth-stage shifts and differences in crop types is also effectively managed, as observed in BIPBIP-2021-Maize $\rightarrow$ NCD and NCD $\rightarrow$ MCD adaptations through masked adversarial alignment on low-confidence pixels, which corrects unreliable pseudolabels in the most domain-divergent regions. Finally, CCUDA maintains consistently high segmentation accuracy even across under represented growth stages through SAS, which is notably demonstrated in the NCD $\rightarrow$ Weedelec Maize and LCD $\rightarrow$ FCD adaptation, thereby ensuring robust generalization across different growth stage distributions.

Collectively, these empirical findings strongly validate our hypothesis: achieving robust cross-crop domain adaptation in agriculture requires domain-specific methodologies explicitly

TABLE I
COMPARING THE PERFORMANCE OF OUR MODEL ON DIFFERENT SOURCE–TARGET DATASETS WITH THE SOTA UDA METHODS.

Source	Target	ADVENT [55]	MINENT [77]	DACS [78]	MUNE [64]	DACS+ [60]	DAFormer [13]	DG-HRDA [11]	HRDA [10]	MIC [12]	CCUDA(ours)	Oracle
Backbone		ResNet-101					Mit-B5					
NCD	BIPBIP 2019 Maize	90.39	90.31	91.56	91.25	92.87	93.14	93.56	93.99	93.99	96.26	97.01
		30.47	37.81	39.02	59.47	41.42	55.87	59.88	60.91	62.68	72.95	86.75
		60.43	64.06	65.29	75.36	68.23	74.50	76.72	77.45	78.22	84.60	91.88
LCD	FCD	89.02	90.1	90.04	95.78	94.12	97.91	94.82	97.98	98.19	98.48	98.86
		29.41	28.14	25.76	69.78	36.45	78.75	63.01	81.38	82.78	85.06	88.74
		59.21	59.12	57.9	82.78	65.28	88.33	78.91	89.68	90.48	91.77	93.8
NCD	WeedElec Maize	90.82	90.47	95.12	98.16	98.45	99.94	99.36	99.87	99.95	99.96	99.97
		29.76	36.45	49.75	73.12	58.45	82.15	61.72	77.66	82.18	87.46	89.55
		60.29	63.46	72.43	85.64	78.45	91.05	80.54	88.76	92.56	93.71	94.75
PEAD Maize	NCD	79.87	83.12	82.74	88.56	86.79	86.75	90.8	89.41	90.56	91.21	94.58
		1.57	1.96	2.57	32.08	11.56	17.67	50.27	37.96	46.02	60.31	68.47
		40.72	42.54	42.65	60.28	49.17	52.21	70.53	63.69	68.29	75.76	81.52
BIPBIP 2021 Maize	NCD	83.79	82.75	85.17	87.26	87.75	86.62	86.7	89.88	92.52	93.1	94.58
		8.48	11.65	14.78	40.89	19.46	16.32	44.43	42.13	61.04	64.69	68.47
		46.13	47.2	49.97	64.24	53.60	51.47	65.57	66	76.78	78.89	81.52
NCD	MCD	95.75	94.37	97.23	98.97	98.17	98.47	98.93	99.17	99.23	99.1	99.59
		35.74	40.78	48.23	50.14	50.75	40.82	57.18	56.53	59.35	61.24	77.57
		65.74	67.57	72.73	74.55	74.46	69.64	78.05	77.85	79.29	80.17	88.58

The metrics used are IoU-Background, IoU-Crop, and mIoU. These metrics are calculated for each source–target combination in the order mentioned above. Boldface highlights the best-performing model and its respective metric values for the target domain. The results are averaged over three data sampling seeds. Bold values indicate the highest performance.

TABLE II
CCUDA APPLICATION ON OTHER SOTA UDA METHODS WITH COST VERSUS BENEFIT ANALYSIS

UDA Method	Backbone	mIoU			Training GPU Memory (GB)			Train Eff. (GB/mIoU)	Inference GPU Memory (GB)			Infer Eff. (GB/mIoU)
		w/o CCUDA	w/ CCUDA	+/-	w/o CCUDA	w/ CCUDA	+/-		w/o CCUDA	w/ CCUDA	+/-	
DAFormer [13]	Mit-B5 [46]	74.50	80.15	+5.65	9.64	10.8	+1.16	0.20	1.00	1.00	+0	0
DG-HRDA [11]	Mit-B5 [46]	76.72	82.04	+5.32	22.46	23.62	+1.16	0.21	9.40	9.40	+0	0
MIC [12]	Mit-B5 [46]	78.22	84.60	+6.38	22.55	23.71	+1.16	0.18	9.40	9.40	+0	0
HRDA [10]	Mit-B5 [46]	77.45	82.94	+5.49	22.46	23.62	+1.16	0.21	9.40	9.40	+0	0

targeting typical sources of model degradation. CCUDA effectively bridges the gap between conventional UDA frameworks and the unique challenges inherent to precision agriculture, delivering substantial and consistent improvements across varied agricultural scenarios.

C. CCUDA Plug-and-Play Characteristic

We evaluate the plug-and-play compatibility of our proposed CCUDA framework by integrating it into four recent SOTA UDA methods, including DAFormer [13], DG-HRDA [11], HRDA [10], and MIC [12]. Table II summarizes the results for the challenging adaptation scenario of NCD → BIPBIP-2019-Maize. The consistent improvements observed across all baseline UDA methods demonstrate the general applicability of CCUDA and that it positively reinforces the domain adaptation process. Specifically, CCUDA substantially boosts mIoU performance, delivering gains ranging from +5.32% to +6.38%.

Among the evaluated approaches, MIC [12] achieves the highest improvement when integrated with CCUDA (+6.38% mIoU). This notable enhancement can be attributed to MIC’s multiresolution image-masking approach, which inherently supports contextual learning. The synergy between MIC’s contextual masking and CCUDA’s targeted refinement of ambiguous regions further amplifies segmentation robustness, thereby explaining the high performance gains. The systematic improvements across diverse SOTA architectures validate CCUDA’s effectiveness in mitigating agriculture-specific domain shifts. Finally, to assess computational efficiency, we introduce the train

and inference efficiency (GB/mIoU) metrics, which quantify the additional GPU memory required per unit mIoU improvement. Across all evaluated methods, CCUDA maintains a low training overhead of around 0.18 to 0.22 GB per mIoU gain, while inference efficiency remains nearly constant at 0.00 GB/mIoU, indicating negligible additional runtime costs. These results confirm that CCUDA achieves substantial accuracy gains with minimal computational overhead, making it highly suitable for real-world precision agriculture applications, where maintaining a favourable balance between performance improvement and resource consumption is essential for scalable and computationally cost-efficient deployment.

D. Error Breakdown and Reliability Analysis

Alongside the standard IoU-based evaluation, we further incorporate accuracy, precision, recall, and AP to provide a more comprehensive assessment of segmentation reliability under domain shift, computed on the target datasets and averaged over three data sampling seeds. As shown in Table III, CCUDA consistently achieves higher mean scores across all metrics compared to the previous SOTA (MIC). The gain in precision is particularly notable, indicating that CCUDA significantly reduces weed-as-crop misclassification under all scenarios while improving the detection of true crop regions, even in ambiguous areas along crop–weed boundaries and within visually cluttered field conditions. The consistently elevated AP across all adaptations further validates CCUDA’s balanced tradeoff between precision and recall, depicting that the model maintains both

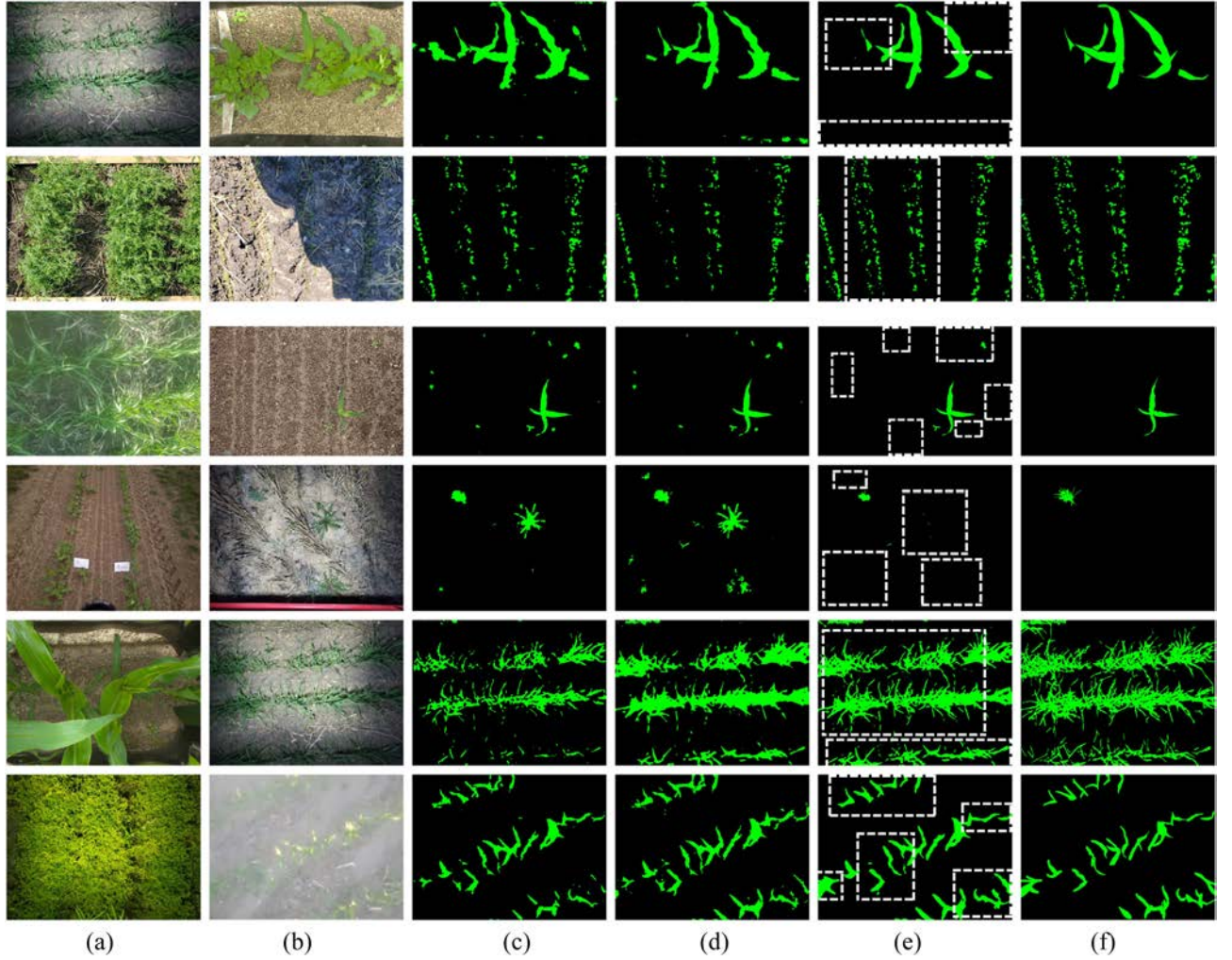


Fig. 2. Qualitative comparison of CCUDA with SOTA UDA methods applied to NCD → BIPBIP-2019-Maize (Row-1), LCD → FCD (Row-2), NCD → Weedelec Maize (Row-3), PEAD Maize → NCD (Row-4), BIPBIP-2021-Maize → NCD (Row-5), and NCD → MCD (Row-6). The dotted rectangles show areas with major improvements in CCUDA's prediction. CCUDA performs better in distinguishing weeds from desired crops, leveraging crop row feature recognition. It demonstrates enhanced effectiveness not only in shadowed or illuminated areas but also in regions where shadows and brightness intersect, as well as in blurred zones. (a) Source. (b) Target. (c) HRDA [10]. (d) MIC [12]. (e) CCUDA(Ours). (f) GT.

TABLE III
ERROR BREAKDOWN AND QUANTITATIVE COMPARISON OF CCUDA WITH PREVIOUS SOTA (MIC)

Source	Target	Method	IoU			Accuracy			Precision(%)	Recall (%)	AP(%)
			background	crop	mean	background	crop	mean			
NCD	BIPBIP 2019 Maize	MIC	93.99±0.04	62.68±0.20	78.22±0.12	96.40±0.03	76.50±0.31	86.45±0.17	76.14±0.25	95.23±0.14	76.91±0.20
		CCUDA	96.26±0.03	72.95±0.13	84.60±0.08	98.20±0.02	84.90±0.12	92.55±0.07	85.35±0.18	97.45±0.09	90.72±0.10
LCD	FCD	MIC	98.19±0.03	82.78±0.65	90.48±0.34	97.53±0.04	83.99±0.38	93.76±0.21	89.45±0.17	95.74±0.11	92.75±0.16
		CCUDA	98.48±0.02	85.06±0.20	91.77±0.11	97.87±0.02	87.09±0.12	96.48±0.07	96.75±0.10	98.34±0.06	97.31±0.08
NCD	WeedElec Maize	MIC	99.95±0.01	82.18±0.53	92.56±0.27	99.90±0.01	88.05±0.33	94.97±0.17	88.47±0.20	99.99±0.02	96.12±0.13
		CCUDA	99.96±0.01	87.46±0.09	93.71±0.05	99.95±0.01	93.50±0.11	97.73±0.06	94.78±0.08	99.99±0.01	98.41±0.05
PEAD	NCD	MIC	90.56±0.08	46.02±0.86	68.29±0.47	94.70±0.05	62.50±0.47	78.60±0.26	62.17±0.31	83.74±0.25	70.44±0.24
		CCUDA	91.21±0.05	60.31±0.31	75.76±0.17	95.20±0.04	79.80±0.18	89.50±0.11	91.26±0.13	94.79±0.10	89.09±0.09
BIPBIP 2021 Maize	NCD	MIC	92.52±0.11	61.04±1.03	76.78±0.57	95.80±0.08	75.10±1.40	85.45±0.74	75.39±0.42	94.32±0.21	83.15±0.36
		CCUDA	93.10±0.05	64.69±0.37	78.89±0.21	96.20±0.03	84.60±0.47	90.50±0.25	94.57±0.12	95.38±0.09	88.77±0.14
NCD	MCD	MIC	99.23±0.02	59.35±0.72	79.29±0.37	99.50±0.01	70.10±0.41	84.51±0.21	70.35±0.26	98.31±0.10	91.92±0.15
		CCUDA	99.10±0.01	61.24±0.23	80.17±0.12	99.55±0.01	76.00±0.15	90.14±0.08	81.79±0.11	99.10±0.04	93.67±0.08

The table reports per-class (background and crop) and mean values for IoU, accuracy, precision, recall, and AP across agricultural source-target pairs, each averaged over three data sampling seeds with their respective standard deviations. CCUDA consistently achieves higher mean scores across all metrics. In particular, its improved precision reflects a stronger ability to suppress weed-as-crop misclassification, while higher AP indicates better maintenance of the precision-recall balance—i.e., CCUDA sustains both prediction confidence and coverage across uncertain crop-weed boundaries. These improvements, coupled with lower variance, demonstrate CCUDA's enhanced model reliability, robustness, and stable cross-domain generalization in agricultural imagery.

high confidence and completeness when segmenting heterogeneous vegetation—accurately identifying crops without missing true positives in complex canopy structures or partially occluded regions.

These improvements are most pronounced in high domain-gap adaptations, such as NCD → WeedElec Maize and PEAD → NCD, where models must generalize from nighttime or blurred imagery to daylight conditions with differing camera setups and

crop phenological stages. The reliability of these findings is further reinforced by the confusion matrices (see Fig. 3), which visually demonstrate a clear reduction in false positives (weed pixels misclassified as crops) and an increase in true-positive detections across all source-target pairs. Likewise, the precision-recall curves (see Fig. 4) reveal consistently increased AP values for CCUDA, confirming its superior ability to sustain prediction confidence across varying thresholds. Collectively, these results

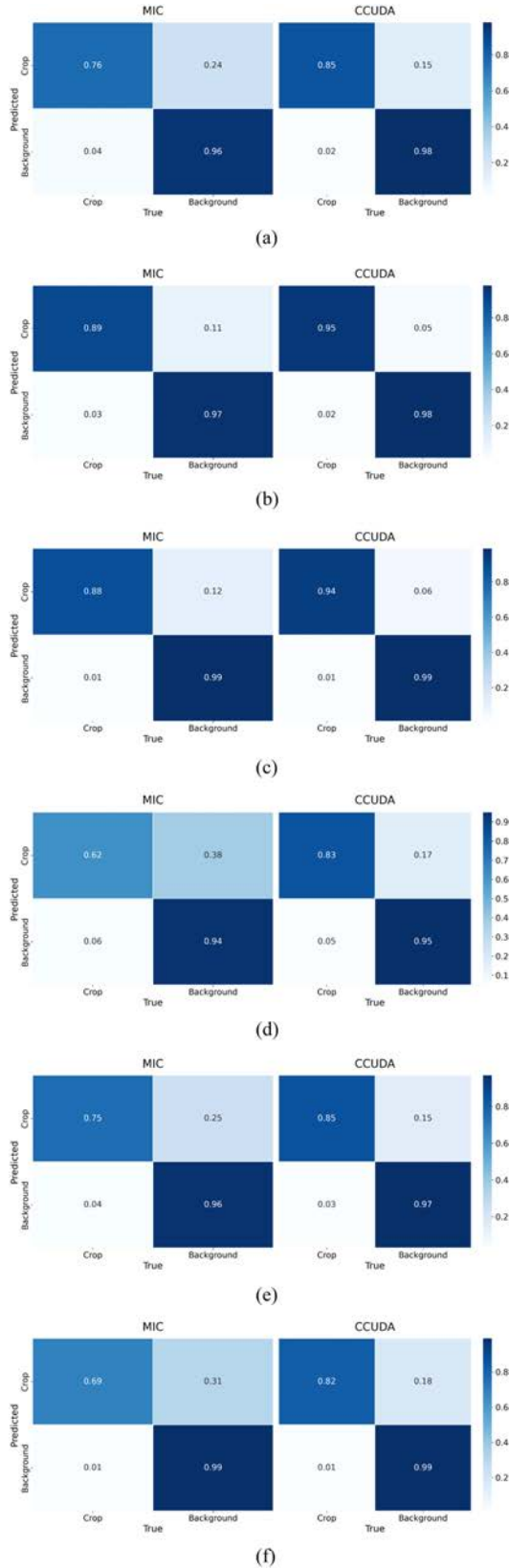


Fig. 3. Confusion matrices across each source–target adaptation computed on the target dataset for MIC and CCUDA, illustrating CCUDA’s reduction in weed-as-crop misclassification and its higher true-positive segmentation accuracy. (a) NCD → BIPBIP-2019-Maize. (b) LCD → FCD. (c) NCD → Weedelec Maize. (d) PEAD Maize → NCD. (e) BIPBIP-2021-Maize → NCD. (f) NCD → MCD.

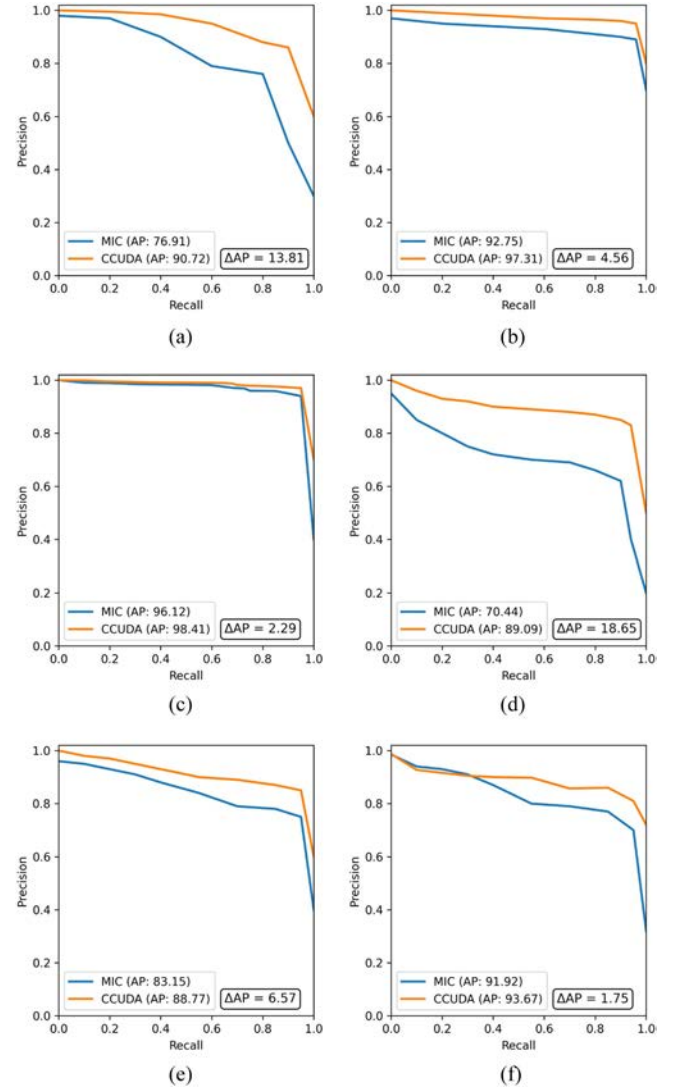


Fig. 4. Precision–recall curves for (a) NCD → BIPBIP-2019-Maize, (b) LCD → FCD, (c) NCD → Weedelec Maize, (d) PEAD Maize → NCD, (e) BIPBIP-2021-Maize → NCD, and (f) NCD → MCD adaptations computed on their respective target dataset for MIC and CCUDA, illustrating CCUDA’s precise balance between precision and recall along with higher AP depicting its superior ability to maintain confident, accurate segmentation under domain shifts while reducing weed-as-crop misclassification.

affirm that CCUDA not only enhances per-pixel segmentation accuracy but also achieves greater reliability, robustness, and stability across diverse and challenging agricultural environments.

E. Ablation Studies

We present a detailed analysis of all components of CCUDA and their impact on the UDA performance in terms of mIoU for NCD → BIPBIP-2019-Maize and PEAD Maize → NCD adaptation scenarios in Table IV. We begin with the HT technique, which improves the average mIoU by +2.5%. It includes the use of masked adversarial loss, which contributes to performance enhancement. As we add other components of CCUDA, we witness further efficiency improvements. One notable component is the LDR technique applied to target images, which

TABLE IV

TABLE PERFORMS AN ABLATION STUDY ILLUSTRATING THE IMPACT OF HT, LDR, AND SAS ON NCD $\rightarrow$ BIPBIP-2019-MAIZE AND PEAD $\rightarrow$ NCD CROSS-CROP DOMAIN ADAPTATIONS IN TERMS OF mIoU (%)

HT	LDR	SAS	MIC [12]	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
X	X	X	✓	78.22 $\pm$ 0.12	68.29 $\pm$ 0.47	73.26
✓	X	X	✓	80.53 $\pm$ 0.14	70.99 $\pm$ 0.51	75.76
X	✓	X	✓	80.06 $\pm$ 0.11	70.23 $\pm$ 0.29	75.15
X	X	✓	✓	79.96 $\pm$ 0.10	70.32 $\pm$ 0.25	75.14
✓	✓	X	✓	82.47 $\pm$ 0.11	73.26 $\pm$ 0.30	77.87
✓	X	✓	✓	82.27 $\pm$ 0.12	72.97 $\pm$ 0.27	77.62
✓	✓	✓	X	82.94 $\pm$ 0.38	73.86 $\pm$ 0.50	78.40
✓	✓	✓	✓	84.60 $\pm$ 0.08	75.76 $\pm$ 0.17	80.18

Each added module contributes to a steady increase in mIoU, demonstrating the cumulative effect of these techniques on enhancing model performance.

TABLE V

EFFECT OF MASKED ADVERSARIAL STRATEGY

Strategy	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
All Pixels	83.03 $\pm$ 0.42	74.18 $\pm$ 0.57	78.61
High-Confidence Pixels Only ($\pi > 0.95$)	82.45 $\pm$ 0.39	73.25 $\pm$ 0.54	77.85
Low-Confidence Pixels Only ($\pi < 0.95$)	84.60 $\pm$ 0.08	75.76 $\pm$ 0.17	80.18
No Adversarial	82.21 $\pm$ 0.28	72.89 $\pm$ 0.31	77.50

Results are reported as mean $\pm$ standard deviation of mIoU across three runs on cross-domain adaptations.

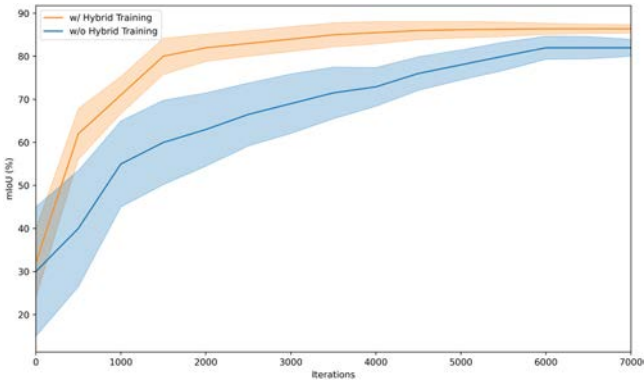


Fig. 5. CCUDA performance on the target domain with and without HT. The curves depict early convergence with the HT.

enhances UDA performance by +1.89%. Similarly, the SAS algorithm improves domain robustness, increasing the average performance by +1.88%. Combining HT and LDR yields even better results, highlighting their complementary nature.

1) *Hybrid Training*: We evaluate the impact of HT throughout the training iterations; accordingly, Fig. 5 plots the evolution of target-domain mIoU on the validation set as pseudolabels are progressively refined with and without HT. The shaded ribbon denotes the standard deviation across three independent data sampling seeds, each with multiple training runs, i.e., the uncertainty band of each curve. As illustrated in Fig. 5, the difference in convergence dynamics is stark. Without HT (blue curve), the model shows slow improvement over time. It takes significantly longer to reach maximum performance levels. The wide uncertainty band, especially during initial iterations, further reflects the high variability and standard deviation across different training runs and random seeds, reinforcing the instability caused by error accumulation due to pseudolabel drift. As early and noisy pseudolabels are repeatedly reinforced during training, they propagate errors that downgrade performance.

With masked adversarial alignment-based HT (orange curve), the model exhibits rapid and stable convergence, in stark contrast to the slow progression seen without HT. The consistently narrow uncertainty band reflects low variance across different data

TABLE VI

EFFECT OF MASKING THRESHOLD β ON ADVERSARIAL REGION SELECTION

β	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
0.50	80.12 $\pm$ 0.86	71.78 $\pm$ 0.95	75.95
0.70	82.12 $\pm$ 0.64	73.42 $\pm$ 0.68	77.77
0.80	83.03 $\pm$ 0.48	74.18 $\pm$ 0.57	78.61
0.90	83.75 $\pm$ 0.43	74.88 $\pm$ 0.52	79.32
0.95	84.60 $\pm$ 0.08	75.76 $\pm$ 0.17	80.18
0.99	83.02 $\pm$ 0.47	73.54 $\pm$ 0.62	78.28

Pixels with confidence (π) $< \beta$ are included in adversarial alignment. Low β values exclude key low-confident regions, while very high β start perturbing confident, well-aligned high-confidence pixels.

sampling seeds, empirically validating suppressed pseudolabel drift and, thus, reducing error propagation.

Building on the convergence behavior shown in Fig. 5, Table V quantifies the impact of different masked adversarial strategies under HT. Aligning features only at low-confidence pixels achieves the highest performance, with an average mIoU of 80.18%, surpassing both full-map and high-confidence-only alignment. This indicates that focussing adversarial adaptation on uncertain regions—where morphology, illumination, and growth-stage gaps are largest—enables the model to correct unstable pseudolabels while avoiding reinforcement of systematic bias. In contrast, leaving well-aligned high-confidence regions (such as row-aligned soil, residue, and background weeds) untouched further suppresses pseudolabel drift, reduces noisy gradient propagation, and stabilizes adaptation across training runs. Applying adversarial alignment indiscriminately to all pixels or only to confident regions slows adaptation and increases variance due to amplified noise in pseudolabel updates.

We further examine the influence of the masking threshold β in Table VI, which determines which pixels—based on their prediction confidence π —are selected for adversarial learning. Pixels with $\pi < \beta$ are considered uncertain and passed to the discriminator, while confident regions ($\pi \geq \beta$) are excluded. At low β values (≤ 0.8), the mask removes a large number of pixels that actually contain domain shifts—particularly those with moderate confidence near crop–weed boundaries or under varying illumination. These regions are critical for cross-domain adaptation. Excluding them leaves the discriminator with insufficient signal to correct feature discrepancies, resulting in weaker alignment and lower mIoU. As β increases toward 0.90, the discriminator gains access to these informative, moderately uncertain regions while still excluding high-confidence pixels that are already well-aligned between the source and target domains (e.g., soil, residue, and background weeds). This selective focus enables stable and meaningful adversarial gradients, yielding the highest performance. Beyond $\beta > 0.90$, the mask begins to include high-confidence regions. This overexposure of stable pixels disturbs well-aligned representations, introducing gradient noise and slightly degrading performance. Thus, $\beta = 0.90$ provides the best tradeoff—targeting domain-variant regions without corrupting the confident structural priors learnt during supervised training.

Finally, Table VII examines sensitivity to the masked adversarial weight λ_{adv} . Larger values (e.g., 0.1) cause instability and higher variance, as overly strong adversarial gradients dominate supervised learning signals. Conversely, very

TABLE VII
SENSITIVITY TO MASKED ADVERSARIAL LOSS WEIGHT λ_{MADV}

λ_{MADV}	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
0.1	80.12 $\pm$ 0.93	71.98 $\pm$ 1.05	76.05
0.01	83.15 $\pm$ 0.51	73.25 $\pm$ 0.63	78.20
0.001	84.60 $\pm$ 0.08	75.76 $\pm$ 0.17	80.18
0.0001	84.45 $\pm$ 0.29	75.18 $\pm$ 0.33	79.81

TABLE VIII
STABILITY ANALYSIS OF LDR WEIGHING FACTOR λ_{LD} ON THE ADAPTATION

λ_{LD}	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
0.1	83.51 $\pm$ 0.44	74.42 $\pm$ 0.63	78.57
0.01	84.60 $\pm$ 0.08	75.76 $\pm$ 0.17	80.18
0.001	83.47 $\pm$ 0.35	74.16 $\pm$ 0.20	80.18

TABLE IX
EFFECT OF LOGIT MARGIN m ON TARGET-DOMAIN ADAPTATION

Margin (m)	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
0.5	83.12 $\pm$ 0.58	74.02 $\pm$ 0.63	78.57
1.0	84.03 $\pm$ 0.42	75.15 $\pm$ 0.48	79.59
1.5	84.60 $\pm$ 0.36	75.76 $\pm$ 0.39	80.18
2.0	84.45 $\pm$ 0.41	75.51 $\pm$ 0.46	79.98
2.5	83.64 $\pm$ 0.55	74.82 $\pm$ 0.63	79.23
3.0	82.87 $\pm$ 0.61	74.10 $\pm$ 0.71	78.48

Moderate margins (1.0–2.0) provide stable and confident boundaries, while very small or large margins weaken pseudolabel quality.

small weights (e.g., 0.0001) weaken domain alignment and reduce adaptation effectiveness. The optimal balance occurs at $\lambda_{\text{MADV}} = 0.001$, where the model attains the highest average mIoU and lowest variance, indicating stable convergence. This configuration allows adversarial cues to bridge domain gaps—particularly those linked to crop morphology differences, etc—without disrupting class-level consistency, validating the effectiveness of confidence-aware hybrid adversarial training.

2) *Logit-Distance Regularization*: Table VIII illustrates the mIoU variation of CCUDA as the weighting factor λ_{LD} for LDR is altered. The results indicate that CCUDA performs optimally with a λ_{LD} value of 0.01. Increasing λ_{LD} to 0.1 results in a 1.09% decrease in the mIoU. Conversely, decreasing the value of λ_{LD} to 0.001 almost eliminates the effect of LDR, resulting in a 1.13% decrease in mIoU.

Next, we examine the effect of varying the margin hyperparameter m in the LDR term in Table IX. Smaller margins ($m < 1.0$) fail to sufficiently separate class logits, allowing ambiguity to persist along boundaries and under high intraclass variation. Performance improves steadily as m increases to 1.5, where the network learns to enforce confident separations in uncertain regions while maintaining adaptability across domains. Beyond $m > 2.0$, the penalty becomes overly aggressive, forcing hard decisions even for naturally ambiguous pixels. This over separation can increase overconfidence, reduce generalization to unseen target conditions, and marginally lower mIoU. The results suggest that a moderate logit gap ($m=1.5$) achieves optimal balance between confidence sharpening and representational flexibility, promoting smoother domain alignment and higher quality pseudolabels.

TABLE X
EFFECT OF TEACHER MOMENTUM α IN EMA UPDATES ON HYBRID UDA PERFORMANCE

Teacher Momentum (α)	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
0.9	83.45 $\pm$ 0.93	74.21 $\pm$ 1.05	78.83
0.99	83.99 $\pm$ 0.51	75.11 $\pm$ 0.63	79.55
0.999	84.60 $\pm$ 0.34	75.76 $\pm$ 0.39	80.18
0.9999	82.94 $\pm$ 0.29	72.78 $\pm$ 0.33	77.86

Moderate momentum ($\alpha = 0.999$) ensures stable yet adaptive pseudolabels, while too small or too large values cause instability or outdated supervision.

TABLE XI
EFFECT OF CONFIDENCE THRESHOLD τ ON PSEUDOLABEL QUALITY AND UDA PERFORMANCE

Confidence Threshold (τ)	NCD $\rightarrow$ BIPBIP	PEAD $\rightarrow$ NCD	Avg.
0.50	71.10 $\pm$ 9.64	63.15 $\pm$ 11.73	77.63
0.70	75.03 $\pm$ 8.52	70.18 $\pm$ 9.57	78.61
0.80	80.45 $\pm$ 1.39	73.25 $\pm$ 1.54	77.85
0.90	84.60 $\pm$ 0.36	75.76 $\pm$ 0.41	80.18
0.95	83.20 $\pm$ 0.44	74.10 $\pm$ 0.49	78.65

Moderate selectivity ($\tau = 0.9$) yields the best balance between pseudolabel precision and spatial coverage.

Fig. 6 depicts the impact of LDR on enhancing model predictions by increasing feature separability, along with confidence and reducing noise. Panel (a) shows the target-domain images, while (b) presents the ground-truth annotations. Panel (c) displays predictions *without* LDR, where higher uncertainty leads to more false positives (e.g., weeds detected as crops). By contrast, panel (e) shows predictions *with* LDR, resulting in cleaner and more accurate segmentation. To visualize uncertainty, we still employ entropy maps: (d) corresponds to the predictions in (c) and (f) corresponds to those in (e). The lower entropy values in (f) indicate the increased confidence achieved after applying LDR, as increasing the feature separability in the logit space allows the model to make confident predictions, which is crucial for reliable crop segmentation in automated agricultural systems.

3) *Analysis of Teacher Momentum (α)*: Table X shows the influence of the teacher momentum α in the EMA update on hybrid UDA performance. It can be seen that the default value of $\alpha = 0.999$ achieves the best overall average mIoU of 80.18% with the lowest variance, indicating stable and consistent pseudolabel supervision.

When α is smaller (e.g., 0.9–0.99), the teacher updates too quickly, closely tracking noisy student gradients. This leads to higher variance across runs (± 0.93 and ± 0.51) and slightly lower mIoU, as unstable pseudolabels degrade consistency during self-training. Conversely, when α is too large (0.9999), the teacher becomes sluggish, updating too slowly to reflect the student's evolving representation. The resulting pseudolabels lag behind the student, weakening adaptation and reducing performance to an average mIoU of 77.86%.

Overall, $\alpha = 0.999$ provides the optimal tradeoff—ensuring smooth teacher updates that filter gradient noise while maintaining responsiveness to new knowledge from the student.

4) *Effect of Confidence Threshold (τ)*: Table XI shows the influence of the confidence threshold τ , which determines the pixel-level reliability used to weight the target loss in pseudolabel learning. A lower threshold (e.g., $\tau = 0.5 - 0.7$) accepts a broad range of pseudolabel pixels to contribute to the loss

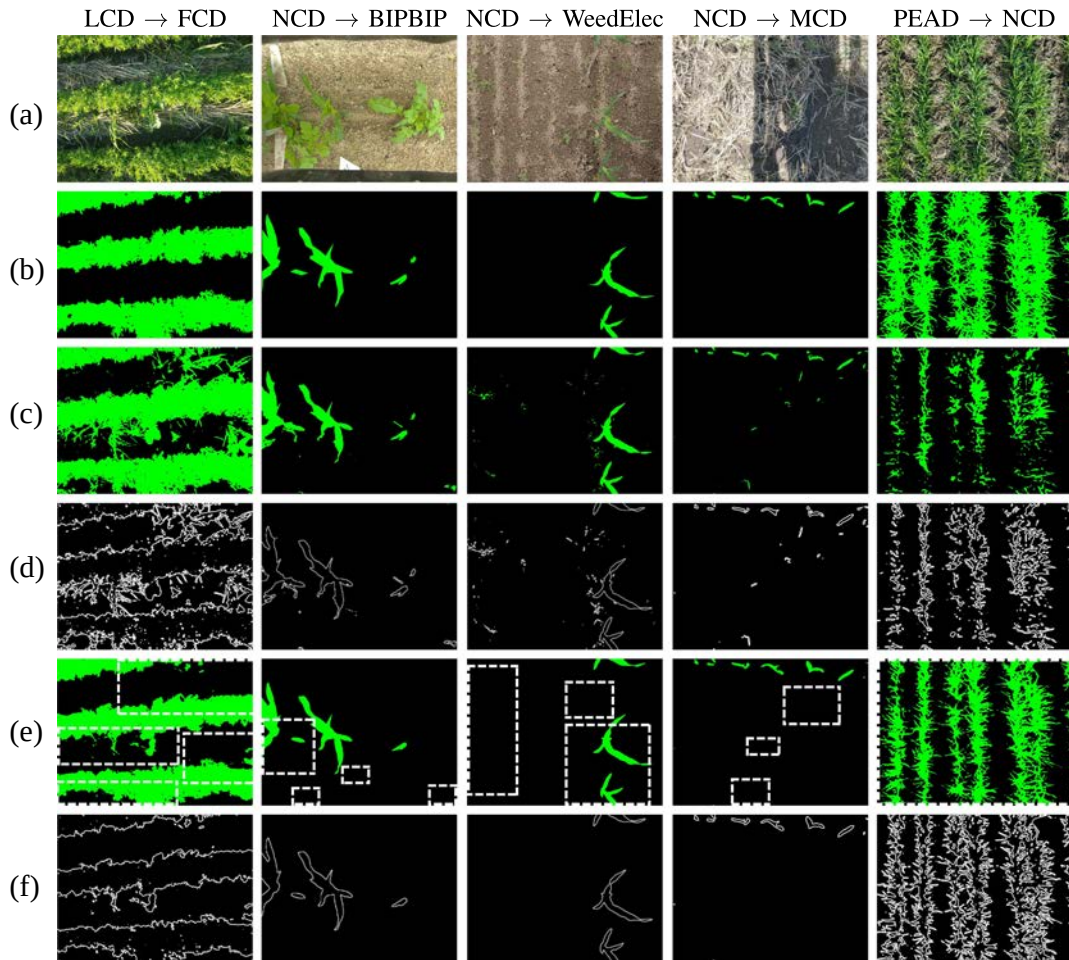


Fig. 6. (a) Target-domain images. (b) Ground-truth annotations. (c) Predictions without LDR. (d) Entropy maps of predictions without LDR. (e) Predictions with LDR. (f) Entropy maps of predictions with LDR.

calculation, including a plethora of low-confidence pixels near class boundaries and in visually ambiguous regions in the field. This increases spatial coverage but introduces noisy pseudolabels, contributing to noisy gradients that lead to higher variance and weaker domain alignment. At the optimal value $\tau = 0.9$, the model achieves the best performance (80.18% mIoU on average) with low variance. This setting effectively filters out unreliable predictions while retaining enough confident pixels for meaningful supervision, yielding stable gradient updates and cleaner pseudolabel propagation across iterations. When τ is raised beyond this level, the pseudolabel coverage drops sharply, leaving too few training pixels from target pseudolabels to guide domain adaptation. Consequently, the model underfits, and convergence slows. These results confirm that moderate selectivity ($\tau = 0.9$) provides the best tradeoff between pseudolabel precision and spatial completeness, minimizing drift while maintaining adaptation strength.

5) *Stage Aware Sampling*: The blue curves in Fig. 7 reveal a pronounced dependence of crop-class IoU on the data sampling seed, indicating that the target domain's dominant growth stage is under represented in the source data. The IoU for crop-class starts increasing at a different rate for different data sampling seeds. We attribute this delay to the sampling order, particularly

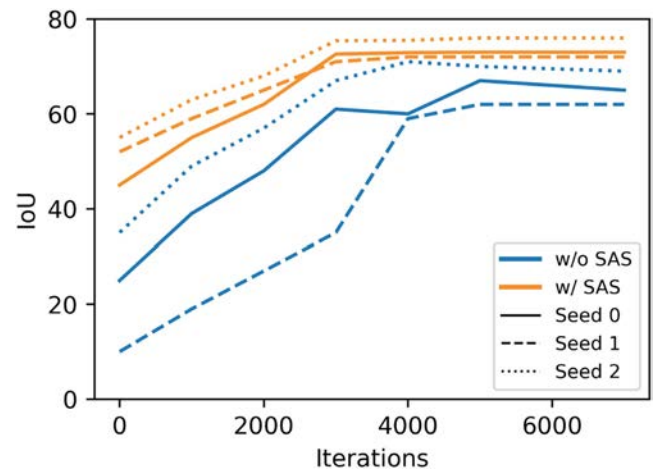


Fig. 7. Effect of SAS on CCUDA performance with varying data sampling seeds.

when the relevant under represented stages are sampled. The later the IoU-Crop acceleration occurs, the lower the final IoU, a consequence of self-training's bias toward the majority stage present in the source domain. Therefore, for UDA, it is essential

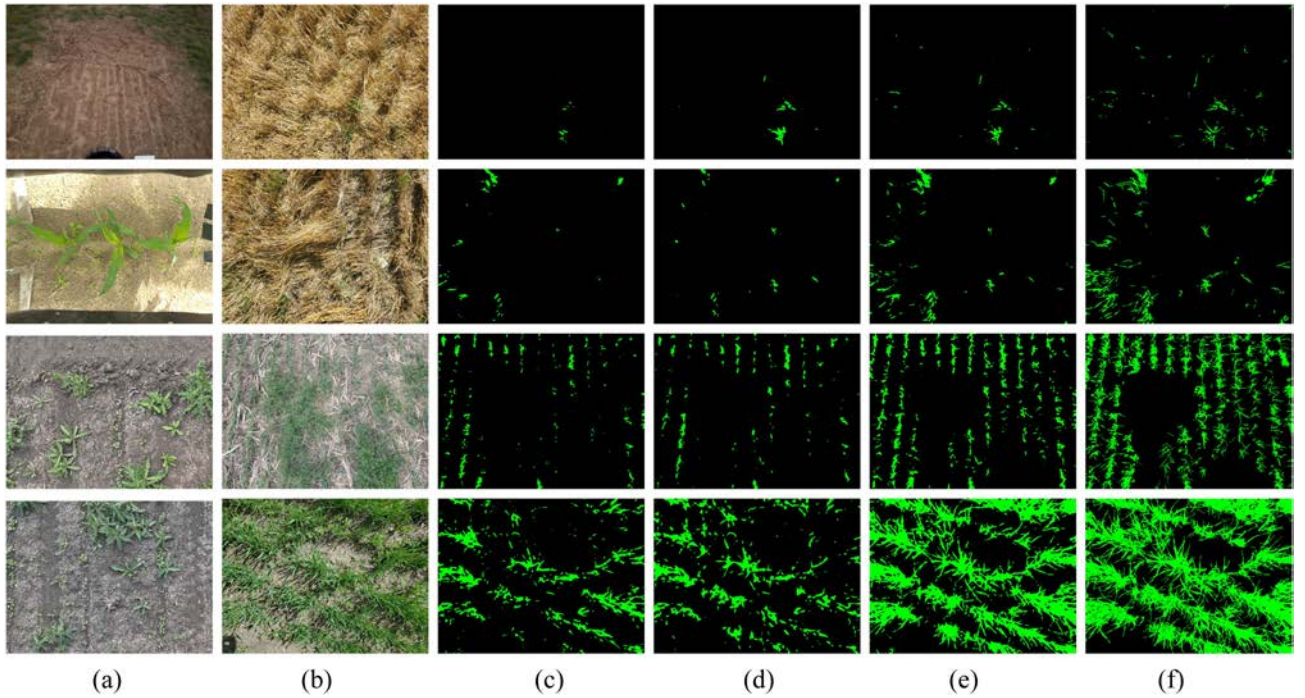


Fig. 8. Failure cases of CCUDA under extreme adaptation conditions with SOTA UDA methods applied to PEAD-Maize $\rightarrow$ NCD (Row-1), BIPBIP-2021-Maize $\rightarrow$ NCD (Row-2), and NCD $\rightarrow$ MCD (Row-3 and 4). Challenging scenarios include complex background residue-based conditions and low image resolution issues, which degrade segmentation accuracy and boundary precision. (a) Source. (b) Target. (c) HRDA [10]. (d) MIC [12]. (e) CCUDA(Ours). (f) GT.

TABLE XII
ABLATION OF THE SEGFORMER ENCODER AND DECODER

Encoder	Decoder	UDA	Oracle
MiT-B5	SegF.	84.10 ± 0.2	91.28 ± 0.1
MiT-B5	DLv3+	82.97 ± 0.8	90.37 ± 0.5
R101	SegF.	74.26 ± 1.4	84.46 ± 1.2
R101	DLv3+	76.32 ± 0.9	87.78 ± 0.9

to learn the under represented growth stage early. SAS addresses this by increasing the sampling probability for the less dominant growth stages, leading to early learning of under represented growth stages and, thus, increased IoU-Crop. This improvement is consistent across different data sampling seeds, indicating that prioritizing the sampling of less dominant stages during training is critical for accurate learning. Fig. 7 also shows that changing the seeds does not affect convergence and maximum IoU with the application of SAS. The efficacy of SAS is evidenced by an increase of almost 7.01% in IoU-Crop.

F. Comparison of Network Architecture

For UDA methods, the transformer-based encoder and decoder have shown promise [13]. To investigate this, we use different encoder-decoder combinations on NCD $\rightarrow$ BIPBIP-2019-Maize. From Table XII, it is evident that the SegFormer-based decoder achieves performance closer to the oracle compared to the DeepLabv3+ decoder. Similarly, the MiT-B5 encoder outperforms ResNet-101. The MiT-B5 encoder and SegFormer decoder are observed to give the highest mIoU relative to the oracle.

G. Failure Cases and Limitations

Although CCUDA achieves consistent improvements across various UDA frameworks, certain challenging scenarios still affect its performance. As illustrated in Fig. 8, when the background soil contains unusually diverse residues along with diverse cross-crop patterns, as in PEAD Maize $\rightarrow$ NCD and BIPBIP 2021 Maize $\rightarrow$ NCD, performance slightly drops due to increased visual ambiguity. Furthermore, when image resolution is intentionally reduced to maintain computational efficiency, fine-grained boundary segmentation becomes less precise due to weakened local contrast and loss of structural detail, as observed in Row-3 and Row-4 of Fig. 8 (NCD $\rightarrow$ MCD). These cases reflect the inherent challenges of extreme visual variability and resolution constraints rather than fundamental limitations of CCUDA. However, it is noteworthy that under these challenging conditions, CCUDA continues to outperform existing SOTA baselines, such as MIC [12] and HRDA [10], demonstrating its robustness and adaptability across diverse domain adaptation scenarios.

V. CONCLUSION

In this article, we have presented CCUDA, a plug-and-play extension to current UDA frameworks that is expressly designed for crop-segmentation under uncontrolled field conditions. CCUDA unifies the following three complementary ideas:

- 1) HT, which alternates supervised learning on source images with teacher-guided self-training on target images while a confidence-aware mask confines adversarial alignment to uncertain pixels, suppressing noisy gradients and pseudolabel drift;

- 2) LDR, which enlarges pixel-wise class margins to curb weed-as-crop mistakes;
- 3) SAS, which balances phenological stages to soften confirmation bias and steady learning dynamics.

Integrated into four state-of-the-art UDA methods, it consistently enhances their performance. As evaluated on eight datasets, CCUDA increases mIoU by up to +5.23% and crop class IoU by +10.04%. These gains underscore the importance of jointly filtering low-confidence pixels, sharpening semantic boundaries, and rebalancing growth-stage distributions, establishing CCUDA as a practical, architecture-agnostic framework for resilient cross-crop domain adaptation.

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